Tableau Assignment

Course: Data Visualization

Tool: Tableau Desktop

1. Objective

Design and implement a multi-source, enterprise-level Business Intelligence system using Tableau. The project must simulate real-world data modeling, advanced analytics, predictive forecasting, cohort analysis, clustering, and performance optimization.

Datasets:

Take the datasets based on your understanding mainly for this task E-Commerce Multi-Table Dataset, Online Retail Dataset, Superstore and Synthetic Marketing Data, etc. datasets you can take it.

2. Dataset Architecture (Mandatory Star Schema)

Students must design a Star or Snowflake schema including:

- Fact Tables:
  - Sales Fact Table
  - Customer Transactions Fact Table
- Dimension Tables:
  - Customer Dimension
  - Product Dimension
  - Geography Dimension
  - Time Dimension
  - Marketing Campaign Dimension

Students must demonstrate logical layer relationships, physical joins, referential integrity, and handling of missing foreign keys.

3. Data Engineering Requirements

- Handle Slowly Changing Dimensions (SCD Type 2 simulation).
- Remove duplicate transactional records.
- Create surrogate keys.
- Compare Live vs Extract connections.
- Explain Hyper Engine compression mechanism.

4. Advanced Analytical Calculations

4.1 Level of Detail (LOD) Expressions

Customer Lifetime Value:

```
{ FIXED [Customer ID] : SUM([Sales]) }
```

Explain FIXED vs INCLUDE vs EXCLUDE and performance impact.

4.2 Cohort Analysis (Mandatory)

- Create acquisition month cohort.
- Build retention percentage matrix.
- Generate survival curve using DATEDIFF and WINDOW functions.

4.3 Statistical KPIs

- 3-Month Moving Average
- Rolling Standard Deviation
- Z-Score Anomaly Detection
- 95% Confidence Interval

Z-Score Formula:

```
(SUM([Sales]) - WINDOW_AVG(SUM([Sales]))) / WINDOW_STDEV(SUM([Sales]))
```

5. Predictive Forecasting

- Use Exponential Smoothing.
- Compare forecast models.
- Calculate MAPE manually:

```
ABS([Actual] - [Forecast]) / [Actual]
```

6. Machine Learning Concepts in Tableau

- Perform k-means clustering for customer segmentation.
- Implement RFM segmentation model.
- Create composite RFM index.

7. Dashboard Design (Three Dashboards Mandatory)

7.1 Executive Control Panel

- KPI Cards with dynamic color coding
- Parameter-driven Top N selector
- Drill-down hierarchy

7.2 Customer Intelligence Lab

- Cohort heatmap
- RFM cluster scatter plot

- Retention curve

7.3 Operational Risk Dashboard

- Anomaly detection chart
- Discount abuse monitoring
- Profit leakage analysis

8. Performance Optimization

- Use Performance Recording.
- Implement Context Filters.
- Compare Data Source vs Extract filters.

9. Theoretical Questions

1. Explain Tableau Architecture.
2. Explain Hyper Engine.
3. Difference between Data Blending and Joining.
4. Explain LOD expressions.
5. Compare Tableau with Power BI and Qlik Sense.

Submission Instructions (Strictly Followed)

Submit **one single PDF or DOCX file only** containing:

- All explanations
- All required screenshots
- Business insights
- `.twbx` file (separately in a folder)
- Raw dataset files (if required)

Important Instructions

- Do NOT submit `.py` or other irrelevant files.
- Do NOT submit incomplete dashboards.
- Ensure all calculations are clearly visible in screenshots.
- Maintain professional formatting and proper labelling.